

Introduction

- Risks concerning **memorization** and extraction of training data: **privacy** and **copyright**.
- **Generative text2image models** are often trained on public data scraped from the internet, e.g. LAION-5B.
- Publicly available data may be subject to copyright.
- **Consent** to use in a specific context is not a consent to use in every context.
- There's a **spectrum** between verbatim copy and completely different.
- The risks grow with the growing popularity of such models.

Our Contribution

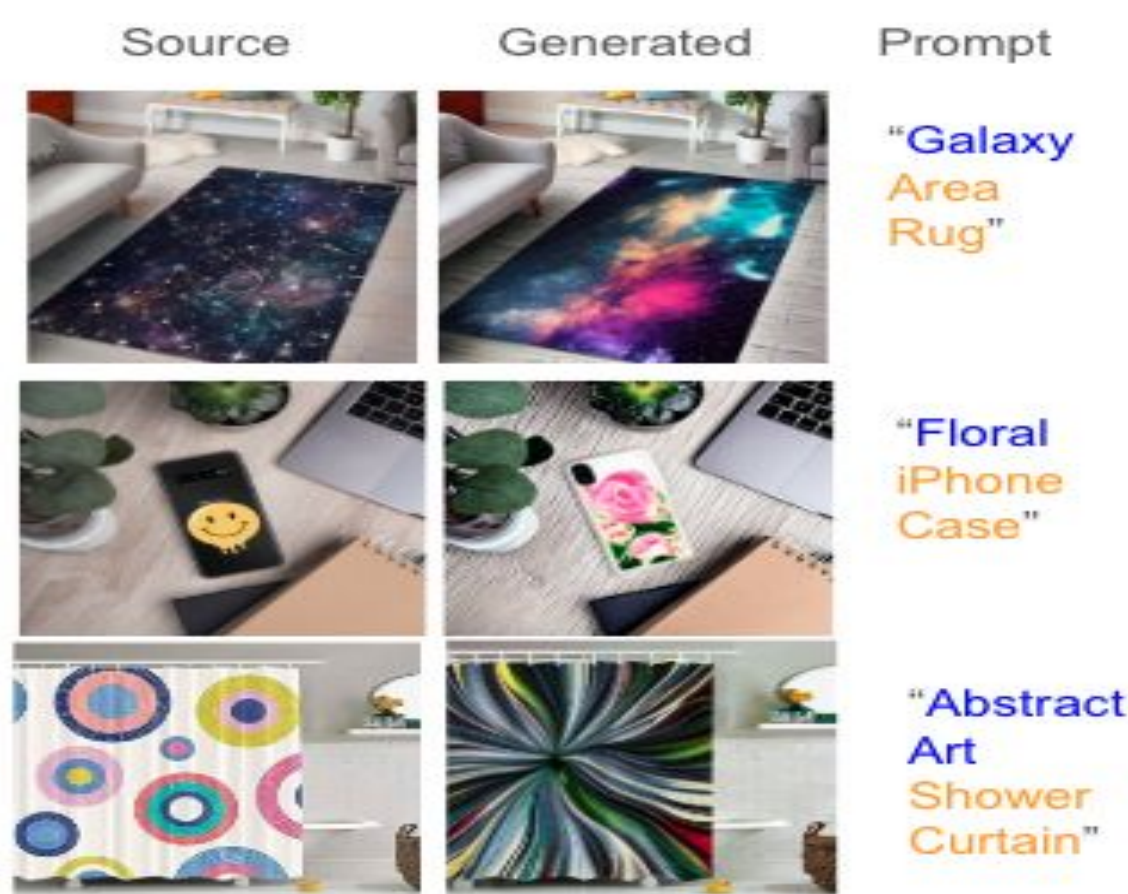
- We developed an attack to extract template memorized images from text2image models.
- Short natural prompts that are likely to be used unintentionally.
- The attack does not require access to the training set or high compute.

Previous Work

- Focused on verbatim memorization.
- Image reconstruction by prompts from training set.
 - Random sampling - Somepalli et al. 2023
 - Duplication in the training set Carlini et al 2023, Webster 2023.

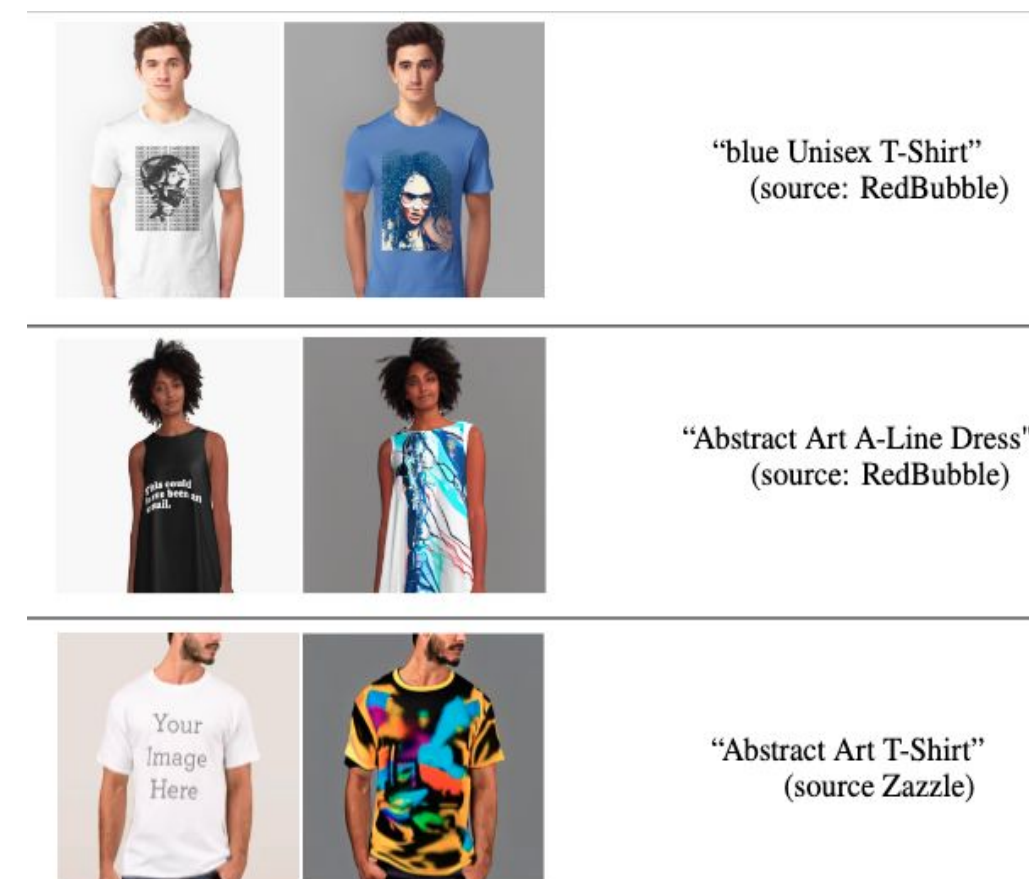
Main Result

Benign prompts successfully reconstruct many template memorized images.

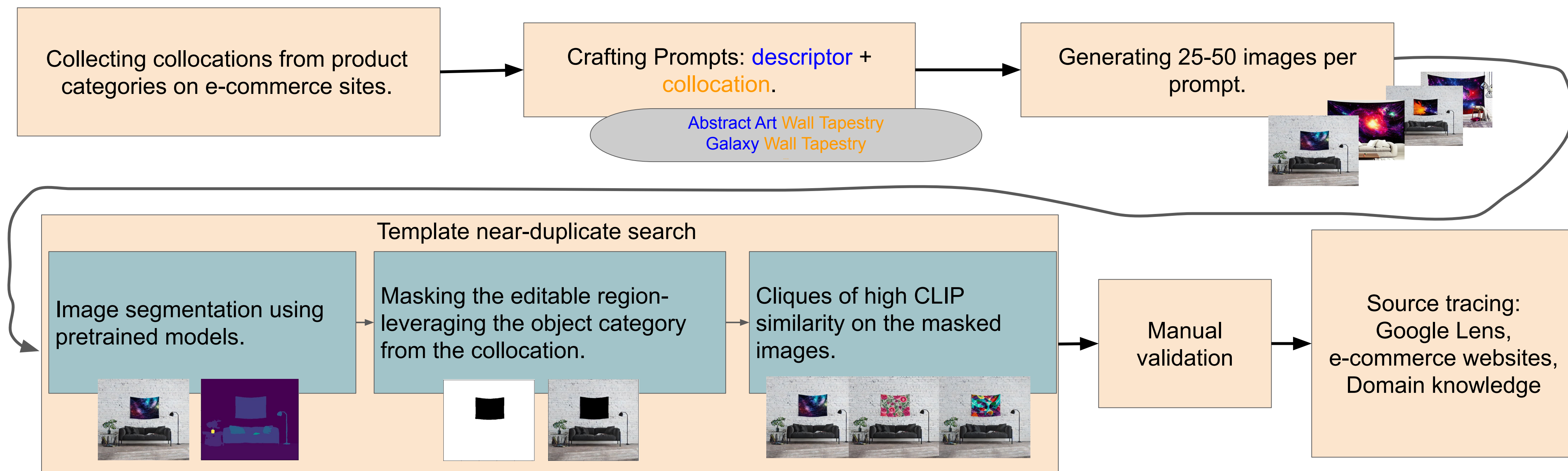


Real People

We extract template-memorized images of real identifiable individuals without specifying their name in the prompt.



Our Attack



Insights Behind Our Attack

- Much of the training data is scraped from e-commerce sites.
- Template image-text structure: a part of the image- the fixed region- is fixed along with a part of the text- the collocation.
- This structure leads to template memorization.



Training set Example from Zazzle.com

Results from SOTA

Even SOTA models are not robust to our method, examples from SD 3.5, right- source.



Observed Phenomena in Template Memorized Images

By observing many reconstructed template memorized images we revealed phenomena never discussed before in literature.

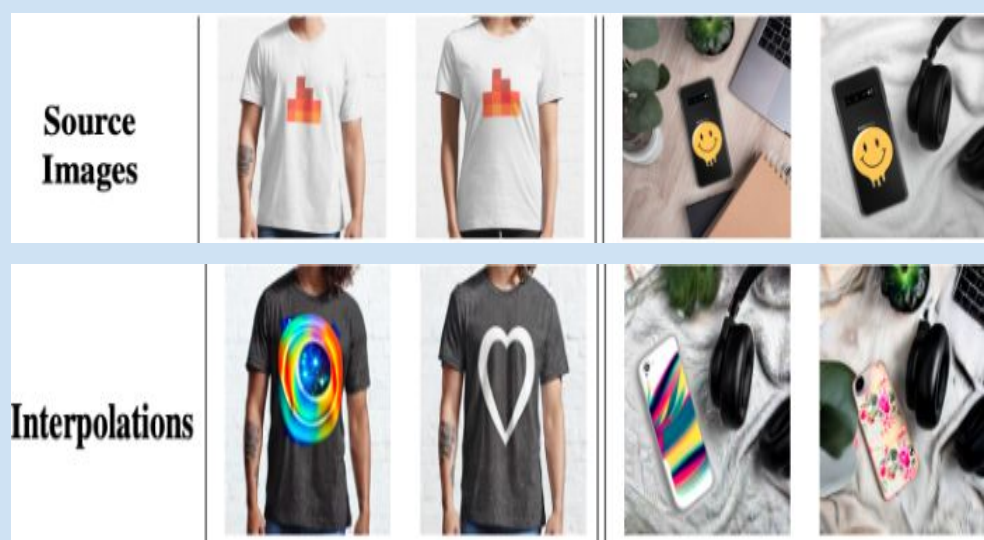
Template Groups Structure

multiple collocations might be associated with multiple image templates. Together we call them a template group. Example collocations: Throw Pillow, Pillow Cover, Pillow Sham.



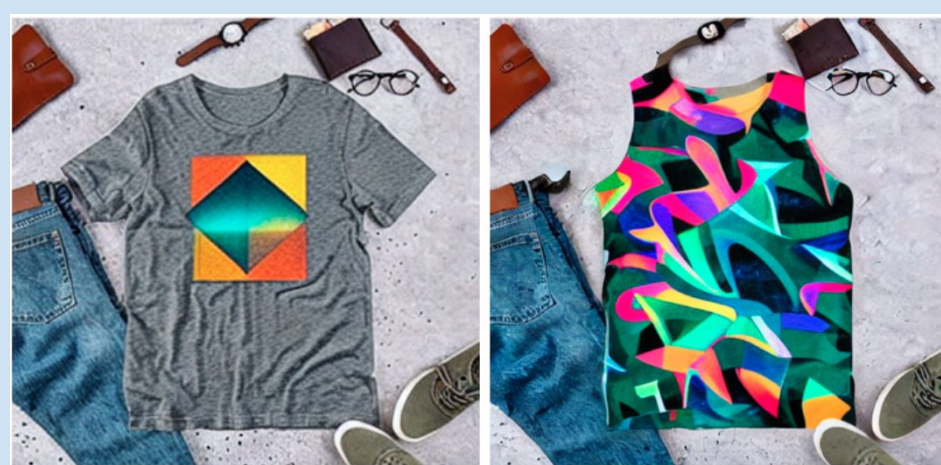
Interpolated Reconstructions

Combination of elements from 2 or more image sources.



Template Leakage.

An object appears with a background memorized for another. Example: T-Shirt -> Tank Top.



Perturbations

Perturbations of objects in the same location and semantical space.

